



上海交通大学

SHANGHAI JIAO TONG UNIVERSITY

Problem F: Cyber Strong?

网络安全政策有效性评估的破局之道

飲水思源 愛國榮校



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飲水思源 愛國榮校



01

真题本质与核心

2025 ICM Problem F: Cyber Strong?



Background:

More and more of our world has become connected through the wonders of modern technology. While this online connectedness has increased global productivity and made the world smaller, it has also increased our individual and collective vulnerability via **cybercrime**. Cybercrime is difficult to counter for a variety of reasons. Many **cybersecurity incidents** cross national borders, complicating issues of jurisdiction for both the investigation and the prosecution of these crimes. Additionally, many institutions, such as investment firms, are unwilling to report a hack, preferring to quietly pay a ransom demand than to let their clients and potential clients know that they were the victim of a security breach. To address the growing cost and risk of cybercrime, many countries have developed national cybersecurity policies, publicly available on their government websites. The International Telecommunication Union (ITU) is the specialized agency of the United Nations focused on information and communication technology; as such, they play a leading role in setting international standards, facilitating international cooperation, and developing assessments to help measure the status of global and national **cybersecurity**.

Requirements:

In this problem, you are asked to help identify patterns that could inform the data-driven development and refinement of national cybersecurity policies and laws based on those that have demonstrated effectiveness. Develop a theory for what makes a strong national cybersecurity policy and present a data-driven analysis to support your theory. In developing and validating your theory, things you may wish to consider include:

- How is cybercrime distributed across the globe? Which countries are disproportionately high targets of cybercrimes, where are cybercrimes successful, where are cybercrimes thwarted, where are cybercrimes reported, where are cybercrimes prosecuted? Do you notice any patterns?
- As you explore the published national security policies of various countries and compare these with the distribution of cybercrimes, what patterns emerge that would help you identify parts of a policy or law that are particularly effective (or particularly ineffective) in addressing cybercrime (through prevention, prosecution, or other mitigation efforts)? Depending on your analytical approach, it may be relevant to consider when each policy was adopted.

- What national demographics (e.g., access to internet, wealth, education levels, etc.) correlate with your cybercrime distribution analysis? And how might these support (or conflate with) your theory?

Based on the quantity, quality, and reliability of the data you collected and used for your analysis, share any limitations and/or concerns that national policy makers should consider when relying on your work to develop and/or refine their national cybersecurity policies.

Your work should not seek to create a new measure of cybersecurity, as there are existing measures such as ITU's Global Cybersecurity Index (GCI),^[1] which assigns a score to each country based on their level of cybersecurity as assessed through five pillars: legal, technical, organizational, capacity building, and cooperation. Instead, you have been asked to seek meaningful patterns in the effectiveness of national cybersecurity policies and/or laws with respect to the national contexts in which those policies were enacted. The GCI or similar existing research may be useful in validating your work. Additional resources that could be useful include websites that collect cybercrime data, particularly those leveraging the VERIS framework, which attempts to standardize how cybercrime data is collected and reported,^[2] including the VERIS Community Database (VCDB).^[3] You are encouraged to find other data sources but be mindful of the veracity and completeness of those sources.

Share Your Insights:

Use your work to create a 1-page memo to country leaders (nontechnical policy experts) attending an upcoming ITU Summit on Cybersecurity. This memo should provide a nontechnical overview of your work, including a summary of the objective and context, your theory, and the most pressing findings that would be relevant to this audience of national policy-makers.

Your PDF solution of no more than 25 pages total should include:

- One-page summary sheet.
- Table of Contents.
- Your complete solution.
- One-page memo.
- Reference List.
- [AI Use Report](#) (If used does not count toward the 25-page limit.)

Note: There is no specific required minimum page length for a complete ICM submission. You may use up to 25 total pages for all your solution work and any additional information you want to include (for example: drawings, diagrams, calculations, tables). Partial solutions are accepted. We permit the careful use of AI such as ChatGPT, although it is not necessary to create a solution to this problem. If you choose to utilize a generative AI, you must follow the [COMAP AI use policy](#). This will result in an additional AI use report that you must add to the end of your PDF solution file and does not count toward the 25 total page limit for your solution.

NEW MCM/ICM: Online Submission Process

The purpose of this article is to assist and guide students and advisors participating in MCM/ICM. In the article, COMAP, provides information about the new online submission process using the new online submission page <https://forms.comap.org/241335097294056>. You will need your team's control number, advisor id number and your problem choice to complete your submission.

References

[1] <https://www.itu.int/publications/publication/global-cybersecurity-index-2024>

[2] <https://verisframework.org/index.html>

[3] <https://verisframework.org/vcdb.html>

Glossary

(The following definitions are derived from definitions provided by multiple International Organizations, including ISO, ITU, and INTERPOL.)

Cybercrime: Cybercrime encompasses a wide range of criminal activities that are carried out using digital devices and/or networks.

Cybersecurity Incident: A single (or a series of) unwanted or unexpected computer security events that have a significant probability of compromising business operations and threatening cybersecurity.

Cybersecurity: Cybersecurity is the collection of tools, policies, security concepts, security safeguards, guidelines, risk management approaches, actions, training, best practices, assurance and technologies that can be used to protect the cyber environment as well as organizational and individual assets.



题面拆解：Background



Background:

More and more of our world has become connected through the wonders of modern technology. While this online connectedness has increased global productivity and made the world smaller, it has also increased our individual and collective vulnerability via **cybercrime**. Cybercrime is difficult to counter for a variety of reasons. Many **cybersecurity incidents** cross national borders, complicating issues of jurisdiction for both the investigation and the prosecution of these crimes. Additionally, many institutions, such as investment firms, are unwilling to report a hack, preferring to quietly pay a ransom demand than to let their clients and potential clients know that they were the victim of a security breach. To address the growing cost and risk of cybercrime, many countries have developed national cybersecurity policies, publicly available on their government websites. The International Telecommunication Union (ITU) is the specialized agency of the United Nations focused on information and communication technology; as such, they play a leading role in setting international standards, facilitating international cooperation, and developing assessments to help measure the status of global and national **cybersecurity**.

背景:

- 随着全球数字化，网络犯罪日益增多，跨国调查与起诉困难，许多机构不愿公开网络攻击事件。
- 各国已制定网络安全政策
- 国际电信联盟（ITU）推动国际合作与评估。



题面拆解: Requirements



Requirements:

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- How is cybercrime distributed across the globe? Which countries are disproportionately high targets of cybercrimes, where are cybercrimes successful, where are cybercrimes thwarted, where are cybercrimes reported, where are cybercrimes prosecuted? Do you notice any patterns?
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Based on the quantity, quality, and reliability of the data you collected and used for your analysis, share any limitations and/or concerns that national policy makers should consider when relying on your work to develop and/or refine their national cybersecurity policies.

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要求:

提出一个理论来解决: 构成强大的国家网络安全政策需要什么? 考虑如下两方面:

1. 网络犯罪的全球分布以及高风险区域

网络犯罪在哪里得逞, 挫败, 被报告, 被起诉?
发现了哪些模式(patterns)?

2. Analyze!

- 有哪些模式(patterns), 能帮助你识别出政策或法律中, 在应对网络犯罪方面特别有效 (或特别无效) 的部分?
- 哪些国家人口统计数据 (如互联网接入情况、财富、教育水平) 与你的网络犯罪分布分析相关?
- 这些数据如何支持 (或混淆) 你的理论?

使用全球网络安全指数 (GCI): 法律、技术、组织、能力建设、合作, 来评估网络安全水平。

还可以使用VERIS社区数据库 (拥有标准化的网络犯罪数据的收集和报告)。



题面拆解：Share Your Insights



Share Your Insights:

Use your work to create a 1-page memo to country leaders (nontechnical policy experts) attending an upcoming ITU Summit on Cybersecurity. This memo should provide a nontechnical overview of your work, including a summary of the objective and context, your theory, and the most pressing findings that would be relevant to this audience of national policy-makers.

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分享你的见解：

- 一页摘要
- 目录
- 完整解决方案
- [一页备忘录](#)
- 参考文献
- AI使用报告

关于备忘录：

- 对象：国家领导人（非技术领域的政策专家）
- 特点：非技术性的概述
- 内容：总结，理论，与这些政策制定者相关的重要发现



本题符合F题的一贯特点：

- 偏文史、管理类专业
- 纯数学方法占比较少
- 需要大量搜集，整合资料

因而，需要我们提炼数据，化繁为简，例如：

如何将全球范围大量网络犯罪数据进行提炼与整合？
怎么选择衡量政策是否有效的指标？



02

我们的探索： 两步入门！

寻找pattern → Analyze!



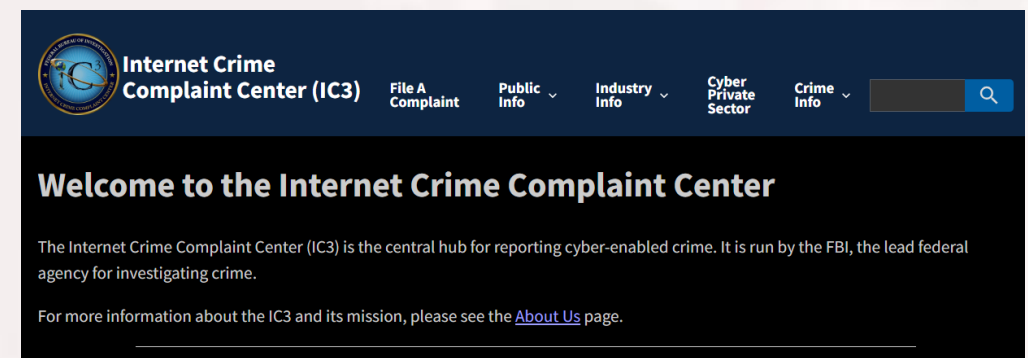
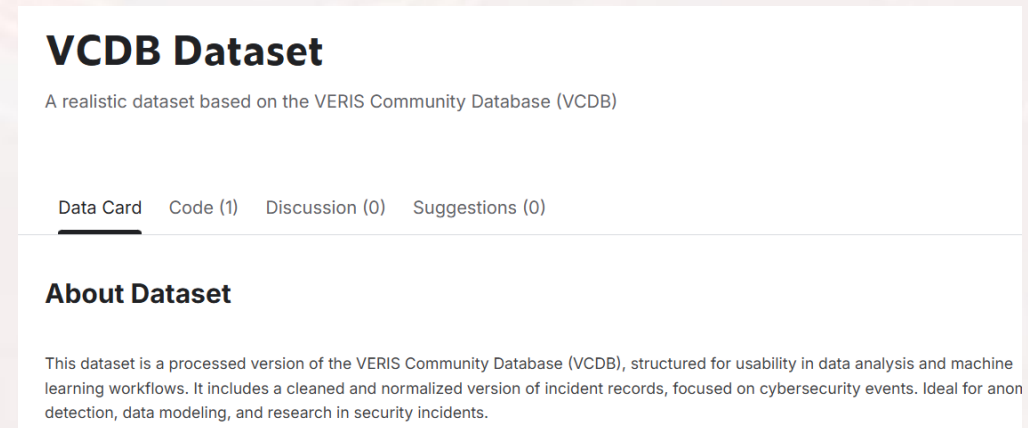
STEP 1:寻找pattern{Requirement 1}



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1.1 找权威数据-对标全球网络犯罪分布

- 电信联盟（ITU）的全球网络安全指数（GCI）
- VERIS的网络安全事件数据库（VCDB）
- 美国互联网犯罪投诉中心（IC3）报告
- （当然国家的白/蓝皮书也可以参考）
- 或者可以在UN相关议题里面查找





STEP 1:寻找pattern{Requirement 1}



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1.2 选取国家样本-对标政策内容、国家背景

可以聚焦**中、美、英、德、日**等5-10个代表性国家

- 覆盖不同经济发展水平与政策体系
- 平衡数据可得性与区域多样性

- （怎么找代表性国家？）

~一个思路：每个洲找2-3个经济体

（发达{如果有}+发展中搭配）

//找易得官方数据的国家也是好方法



STEP 1:寻找pattern{Requirement 1}



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1.3 政策量化:

简易流程

例如[借助NLP](#)

- 通过“**政策文本采集**→**关键词提取**→**词频统计**→**情感倾向评分**”四步流程
- 将非结构化政策文本转化为**可分析数据**
- 适用于大规模政策比较
- **找到可以量化的维度！！**
- （当然用别的更高级的方式也可以

找准维度

对于某个具体政策：

- 可以从法律完备性、技术标准、组织保障、能力建设、国际合作五个维度进行1-5分制打分
- 形成结构化政策强度指数
- 适合数据有限场景。

[然后可以各种可视化！！](#)



STEP 2: Analyze!{Requirement 2}



“In this problem, you are asked to help identify patterns that could inform the data-driven development and refinement of national cybersecurity policies and laws based on those that have demonstrated effectiveness.”

<- 用数据说明政策有效性

2.1 入门级：双变量相关性分析

- 采用 Pearson（线性相关）或 Spearman（秩相关）系数，分析政策量化得分与犯罪率的基础关联
- 用一些简单的方法**初探变量关系**。

Pearson系数

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

Spearman系数

1. **求秩**：分别对变量X和Y的观测值从小到大排序，并记录每个数据对应的秩（排名）。若有数据值相同，则取它们排名的平均值作为共同的秩^①。
2. **计算秩差**：对于每一对观测值，计算其对应的X的秩和Y的秩之差，记为 d_i 。
3. **计算系数**：将秩差代入公式。在没有重复秩的情况下，常用公式为：

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

其中， n 是观测值的对数^{⑥ ⑦}。

4. **结果解读**：系数的取值范围在-1到1之间。正值表示正单调相关，负值表示负单调相关。绝对值越接近1，单调性越强^③。



STEP 2: Analyze! {Requirement 2}



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2.2 进阶级：多元线性回归模型

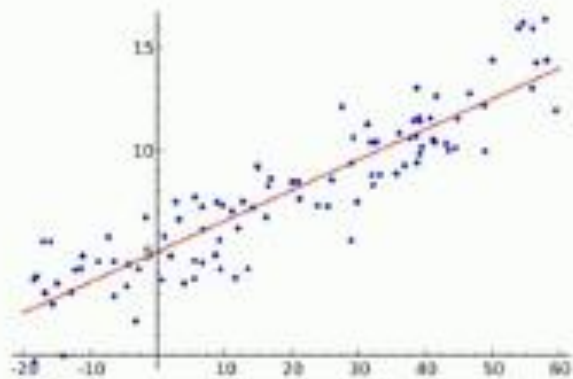
“因变量的变化可由多个自变量的线性组合来解释”

- 以本道赛题为例，可以纳入GDP、互联网普及率等控制变量（在第一步选择出来的指标），构建形如

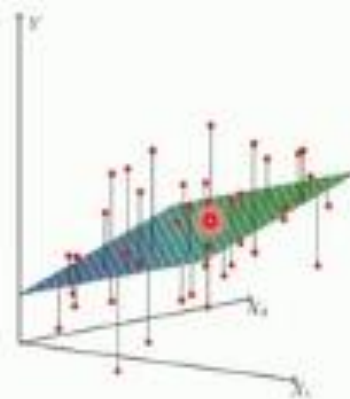
<进一步思考是否仅有政策影响犯罪率？还有哪些可能因素？>

“犯罪率 = 政策强度 + 经济水平 + 技术投入 + 误差项”

的回归方程，量化单一政策因素的独立影响。



一元线性回归



二元线性回归



STEP 2: Analyze! {Requirement 2}



<让机器去猜>

{因为可能涉及人解不出来的方程但是机器可以预测其规律}

2.3 再高一级：机器学习预测模型

在本题中，常见的可能用到的模型有：

- 随机森林模型：可捕捉**非线性关系**与交互效应
- 时间序列预测：适用于政策实施后的**趋势推演**

随机森林的运作机制可以概括为以下三步：

1. **Bootstrap抽样（数据的随机性）**：从原始训练集中**有放回地**随机抽取N个样本（这意味着同一个样本可能被多次抽中），形成一个用于训练单棵决策树的子数据集。该过程会重复进行，为森林中的每一棵树生成一个略有不同的训练集 ① ⑥。
2. **构建决策树（特征的随机性）**：对于每一棵决策树，在每一个节点进行分裂时，并不是评估所有特征来选择最佳分裂点，而是**先随机选取一个特征子集**（通常特征子集的大小设为总特征数的平方根），然后仅在这个子集中寻找最优分裂特征和点 ① ⑥ ⑦。每棵树都会完全生长，一般不进行剪枝。
3. **集成预测**：
 - **分类问题**：当需要对新样本进行分类时，每棵决策树都会独立地给出一个预测类别。随机森林的最终预测结果是所有决策树预测的**众数**（即得票最多的那个类别） ① ③。
 - **回归问题**：对于回归任务，每棵树会预测一个具体数值。随机森林的最终预测结果是所有树预测值的**平均数** ③ ⑨。

这种“双重随机性”的机制是随机森林成功的关键。它确保了森林中的每棵树都各不相同且具有一定独立性，从而有效降低了模型整体过拟合的风险，并增强了泛化能力 ⑦。

常见时间序列模型

（具体原理可以查询相关资料）

ARIMA

通过差分将不平稳的序列转化为平稳序列，然后结合自回归 和移动平均来捕捉数据中的规律

LSTM

长短期记忆网络（LSTM）是一种特殊的循环神经网络（RNN），专为处理和预测时间序列数据中具有长期依赖关系的重要事件而设计。它通过独特的“门控机制”有效克服了传统RNN在训练过程中遇到的梯度消失或爆炸问题。



03

不同等第论文对比



O奖 Five Elements Illuminate, Cybersecurity Innovates

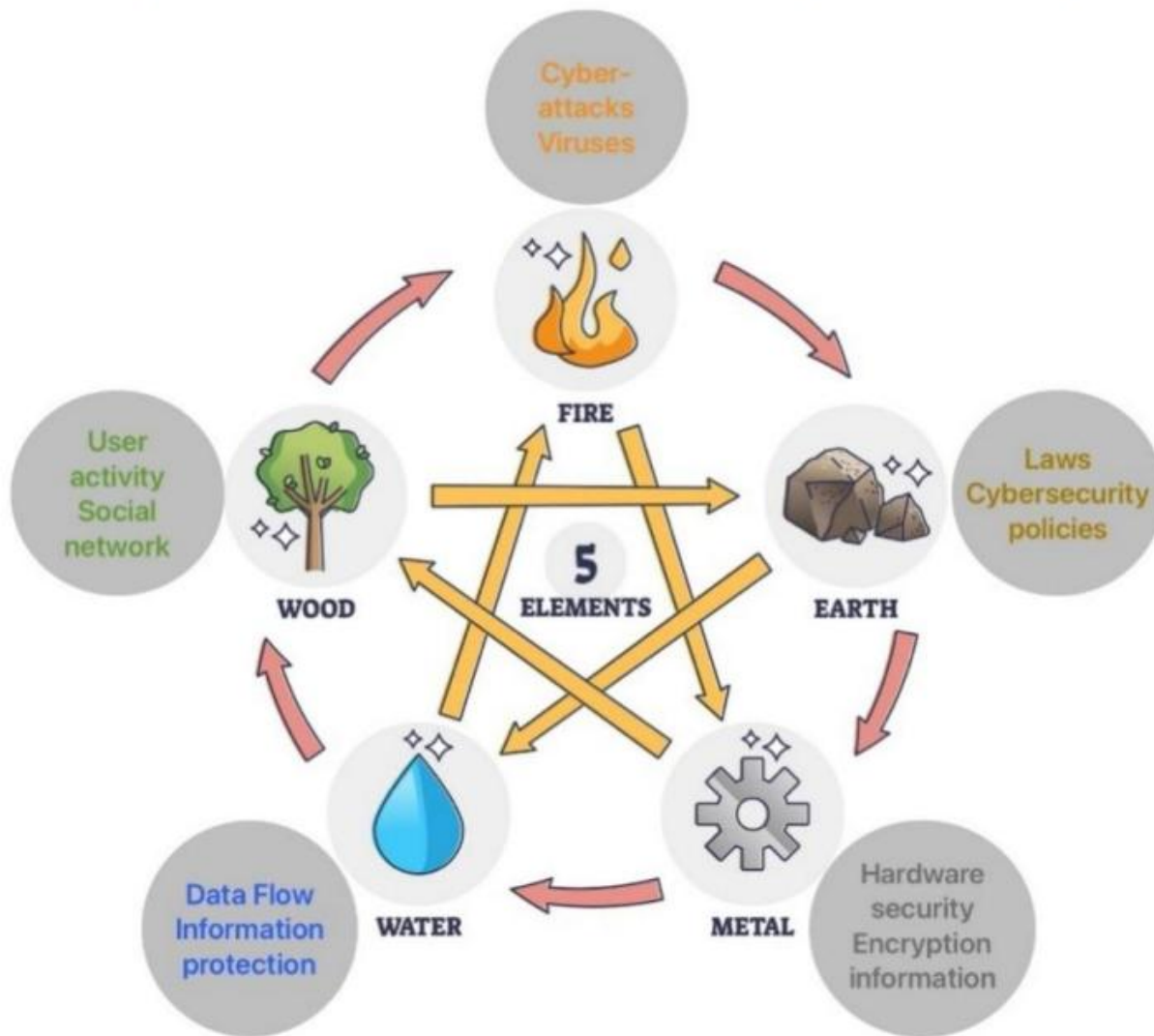
论文标题与核心定位:

[..\2025F-O.pdf](#)

中国传统五行理论

——极具特色!

Figure 2: Generating and Restricting Relationship





O奖 Five Elements Illuminate, Cybersecurity Innovates

模型分析

模型创新融合点

创新性融合五行理论、K-means聚类、差异中的差异（DID）和结构方程模型（SEM），从多维度对网络安全政策进行分析，其中五行理论用于网络安全元素分类。

技术细节呈现

模型构建较为复杂，包含如结构方程

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

等数学公式，但由于使用虚拟数据，一定程度上降低了模型的可靠性。

横向与纵向模型比较

横向比较，与F级论文相比，理论框架更独特但实用性较弱；与H级论文相比，模型系统性更强但数据支持不足。纵向比较，模型从简单聚类演进到多方法融合，但2020年数据未更新，时效性受限。



模型分析

[..\2025F-F.pdf](#)

模型多样性体现

结合负二项回归（处理计数数据，公式：

$$\log(\mu_i) = \beta_0 + \beta_1 X_i$$

随机森林（用于特征重要性分析，如识别GDP影响）和TimeXer（时间序列预测），构建多方法融合的分析框架。

技术细节特点

数据预处理严谨，采用Z-score标准化方法提升数据质量，但模型假设存在简化，如默认单向因果关系，可能对复杂现实情况的刻画不够全面。

横向与纵向模型比较

与O奖相比，模型创新性稍低；与H奖相比，方法体系更全面，综合运用多种分析工具提升研究深度。

模型覆盖2019-2023年数据，能够捕捉短期趋势变化，但对长期效果的分析不足。



M奖 H奖 S奖分析

模型分析

M奖

[..\2025F-M.pdf](#)

- 模型综合热图、NLP、决策树回归和神经网络，NLP用于政策分类，决策树用于特征重要性识别（如安全服务器数）；但神经网络赞助商分类准确率低（31%）， R^2 值较高（0.75-0.78），过拟合风险大。

H奖 (CyberPET)

[..\2025F-H.pdf](#)

CyberPET 模型分“Vision System”（数据可视化）和“Brain System”（宏观-微观分析），宏观用相关性分析，微观用显著性分析，K-means++聚类识别代表性国家；但敏感性分析显示模型稳健性弱，验证不足。

S奖

[..\2025F-S.pdf](#)

模型主要用线性规划和随机森林，对比后优选随机森林，公式简单（如线性回归），依赖GCI和WCI指标，权威但应用窄； $R^2=0.75$ ，但黑箱特性使解释性弱，基于2018-2020年数据，过时且未更新，实用性低。

M奖 Modeling Cybercrime : Distributions, Influential Factors, and Policy Implications

模型分析 [..\2025F-M.pdf](#)

模型多样性体现

- 模型综合热图、NLP、决策树回归和神经网络

- NLP用于政策分类，决策树用于特征重要性识别（如安全服务器数）

- 但神经网络赞助商分类准确率低（31%）， R^2 值较高（0.75-0.78），过拟合风险大。

H奖 CyberPET: Policy Effectiveness Tracking for Global Cybersecurity Governance

模型分析 [..\2025F-H.pdf](#)

模型多样性体现

- CyberPET 模型分“Vision System”（数据可视化）和“Brain System”（宏观-微观分析）

- 宏观用相关性分析，微观用显著性分析

- K-means++ 聚类识别代表性国家；但敏感性分析显示模型稳健性弱，验证不足

模型分析 [..\2025F-S.pdf](#)

模型多样性体现

- 模型主要用线性规划和随机森林，对比后优选随机森林，公式简单（如线性回归）
- 依赖 GCI 和 WCI 指标，权威但应用窄
- $R^2=0.75$ ，但黑箱特性使解释性弱，基于 2018-2020 年数据，过时且未更新，实用性低。

不同奖项论文语言风格比较

语言风格比较

O	F	M	H	S
学术性强，混合中英文术语兼具专业深度与一定主观色彩。	采用学术报告风格，避免过度抽象，政策建议部分用词实用（如“加强经济安全”）整体语言风格易于理解，贴近读者。	学术性重，同时可 视觉化描述用词生动（如“词云”）在保持学术严谨性的基础上，通过生动表述增强内容呈现效果。	技术性高，但数据描述部分用词不精确（如“数据缺失严重”），语言风格偏向技术阐述，但细节表述存在不足。	采用报告式语言，过于简略，政策分析部分用词笼统，整体语言风格缺乏深度和细节，表述较为刻板。



04

经验总结



赛题特点与趋势分析

01

数据驱动+模型建构+政策建议三位一体

- 需从多源公开数据库提取数据（ITU，VERIS等）
- 需建立可解释可评估的数学模型
- 结论需转化为具体可操作的政策建议

02

强调跨学科的融合&创新性

- 五元素模型，CyberPET框架
- 创新也包括分析视角与解释框架

03

可视化与写作能力与评分相关

- 图表需清晰美观信息量大
- 需具备多层次可视化能力（宏观分布->微观特征）
- memo写作能力直接影响模块得分



常见失误

01 数据问题

- ✗ 数据来源单一、未经清洗
- ☑ 多源验证、缺失值处理、标准化说明

02 模型堆砌与解释不足

- ✗ 模型多但无逻辑串联、结果未解释
- ☑ 模型选择理由明确、结果有业务含义解读

03 假设不合理或未说明

- ✗ 假设过于理想化、无合理性说明
- ☑ 每个假设附带“Justification”

04 可视化混乱

- ✗ 图表过多、无标注、配色杂乱
- ☑ 一图一结论、标注清晰、风格统一

05 Memo写得太技术或太泛

- ✗ 使用专业术语、无具体建议
- ☑ 语言通俗、建议可操作、结构清晰

06 参考文献格式混乱

- ✗ 引用不规范、来源不明确
- ☑ 使用标准引用格式、注明网址/报告名称



01

建模建议：

- 从简单模型开始（如相关性分析）
- 逐步深入（聚类 → 回归 → 预测）
- 模型结果要可解释、可可视化

02

写作建议：

- 结构清晰：摘要 → 引言 → 模型 → 结果 → 结论 → Memo
- 语言简洁：避免冗长句子、多用图表代替文字
- 重点突出：每部分开头说明“本部分目标”



<memo>{TASK}



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- O
- F
- M
- H
- S



时间管理与分工建议

01

时间分配（四天为例）：

- day 1: 理解题目、收集数据、初步分析
- day 2: 建模与初步结果
- day 3: 结果深化、写作初稿
- day 4: 修改、优化、整理提交

02

分工建议：

- 1人负责数据收集与清洗
- 1人负责模型构建与实现
- 1人负责写作与图表制作
- 全员参与模型讨论与结果解读



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感谢观看

—— 飲水思源 愛國榮校 ——



NLP!! <自然语言处理>

政策文本采集→关键词提取→词频统计→情感倾向评分

- 政策文本采集：（前两步已经完成啦）

- 关键词提取:(可以借助python实现)

事前准备：建立自定义词典，预设好关键词{比如cybersecurity, infrastructure, incident response, public-private partnership, encryption, critical infrastructure...}

ps:可以在阅读题目时顺便记下，说不定就用到了^^

python处理：中文可以用jieba（但英语论文最后还是要翻译回来）

英文可以用nltk或spaCy

- 词频统计

python统计词频

先做大清洗！（去掉一些坏词{比如不停不停出现的security}）

再呈现词频统计（dataframe模块做出表格{政策-关键词矩阵}）

最后可视化（生成词云即可）//高中信息技术（？

- 情感倾向评分{strength/certainty}

借助已有的词典{TextBlob/ VADER}来进行倾向分析

//（不太建议新创一个捏{从0开始是比较耗时间的}）

- 有什么用？？

相关性分析~（比如看情感与政策效果之间是否有联系）

[RETURN~](#)



可视化结果！！（以热力图为例）

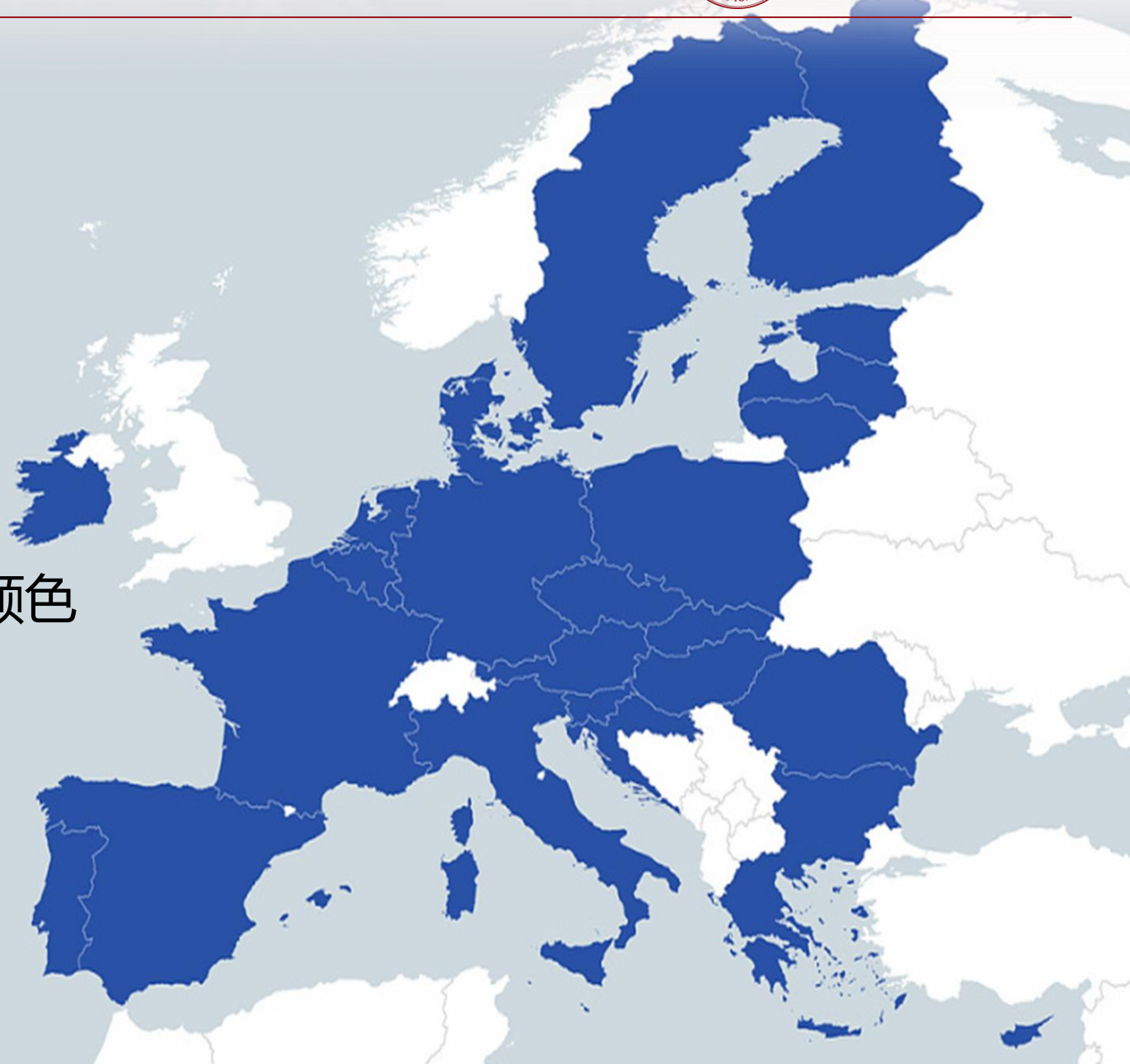


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热力图绘制要点

- 可以使用Python的Geopandas库
- 通过分级设色展示区域差异
- 需注意投影方式选择与异常值处理
(或者使用MATLAB也比较方便)
- 热力图很直观，有震慑力（如果选的颜色恰当的话）

[RETURN~](#)





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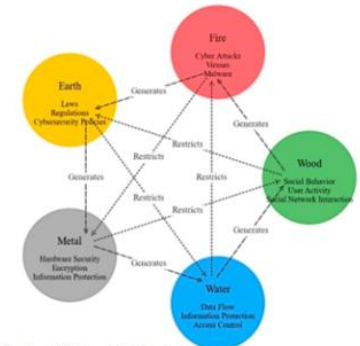
Memo

To: ITU Cybersecurity Summit Attendees

From: MCM Team

Subject: Five-Element Cybersecurity Framework for Policy Innovation

Date: January 25 2025



1. The Rising Cybersecurity Challenge

Cybercrime is a **global security and economic threat**, with projected **\$10.5 trillion USD** in damages by 2025. Nations with **weak laws, poor data governance, and outdated cybersecurity policies** are at higher risk. Without **data-driven, adaptive policies**, ransomware, financial fraud, and cyber espionage will continue to escalate.

2. The Five-Element Cybersecurity Framework

We propose an innovative **Five-Element Model**, integrating policy, technology, and human behavior into a **balanced security ecosystem**:

3. Key Findings & Policy Recommendations

Element	Cybersecurity Domain	Policy Action
Metal (Infrastructure Security)	Encryption, network resilience	Enforce hardware security & encryption standards
Wood (Human & Social Risk)	Phishing, misinformation	Launch nationwide cybersecurity awareness campaigns
Water (Data Protection & Flow)	Data privacy, cloud security	Strengthen cross-border data-sharing & GDPR-like regulations
Fire (Cyber Threats & Attacks)	Malware, hacking, AI-driven threats	Develop AI-powered cyber defense systems
Earth (Laws & Policies)	Cybersecurity laws, global cooperation	Align laws with ITU guidelines & enhance international treaties

Countries with strong cybersecurity laws experience 35% fewer cyberattacks—strengthen **ITU-backed legal frameworks** to enforce national cybersecurity standards.

Global cooperation reduces cross-border cybercrime by 40%—expand **international data protection agreements** to improve cross-border security.

Investing 0.3% of GDP in cybersecurity reduces financial losses by 25%—allocate resources to **AI-powered threat detection** for real-time cyber defense.

Public awareness campaigns lower phishing scams by 28%—launch **nationwide cybersecurity education programs** to reduce social engineering risks.

Enhance international law enforcement collaboration to prosecute cybercriminals effectively and close jurisdictional loopholes.



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MEMORANDUM

To: Country Leaders Participating in the ITU Summit on Cybersecurity

From: ICM Team

Subject: Key Insights on Strengthening National Cybersecurity Policies

Date: January 28, 2025

Esteemed Leaders in Attendance:

With the increasingly severe cybersecurity situation, it has become particularly crucial to strengthen the construction of cybersecurity policies in various countries. We conducted an analysis aiming to identify the factors influencing cybersecurity of different countries, exploring effective cybersecurity policy models, and provided insights to countries on how to enhance their capabilities to combat cybercrime.

Our goal is to analyze the global distribution of cybercrime and the existing policies of various countries to identify the factors influencing cybersecurity, explore successful cybersecurity strategies, and help nations develop more effective cybersecurity policies.

Economically developed countries with widespread internet access are often primary targets of cybercrime. However, nations with robust legal frameworks and strong investments in technology are better equipped to prevent cybercrime. Comprehensive legal frameworks, technological infrastructure, and international cooperation are key factors in enhancing the effectiveness of cybersecurity policies. Countries that update their policies in line with international standards tend to perform better.

There is a strong correlation between internet access, wealth levels, education, and the distribution of cybercrime. Notably, wealthy and well-educated countries often have stronger cybersecurity defenses. Strengthening the legal framework for cybersecurity and increasing investment in technology are essential for improving a nation's cyber defense capabilities. As cybercrime is a transnational issue, enhancing collaboration and data sharing with international partners can significantly improve the ability to combat and prevent cybercrime. Additionally, it is crucial to ensure that citizens and organizations are aware of the risks of cybercrime and are provided with effective protective measures.

Although cybersecurity policies have been gradually strengthened, it is crucial to recognize the challenges posed by cybersecurity. If policies do not evolve continuously, they risk falling behind in addressing increasingly sophisticated cybercrime methods, transnational crimes, and rapid technological advancements. The lag and difficulties in implementing cybersecurity policies remain significant challenges for many countries. Notably, during the outbreak of cybercrime in 2013, policies and regulations failed to respond effectively and promptly, highlighting the need for ongoing development and improvement of cybersecurity policies.

We do hope these insights can provide strong support for your decision-making and collectively contribute to the advancement of cybersecurity across nations.



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Memo

Date: January 28, 2025

To: ITU Summit Attendees (Country Leaders, Policy Experts)

From:

Subject: Strengthening Cybersecurity through

Policy Evaluation and Demographics Analytic



Dear National Leaders,

Cybersecurity is a top priority in today's digital world, requiring strong strategies to reduce risks and build resilience. Our team has created a framework to evaluate cybersecurity policies and a predictive model to assess cybercrime incidents and financial losses. Below are our key findings and recommendations to support better policy-making.

1. Effective Cybersecurity Policies:

We assess cybersecurity policies across five key dimensions: **comprehensiveness, strength, innovation, cooperation, and public participation**. Our research highlights the importance of policies that **take a holistic approach, enforce robust measures, encourage innovation, and foster international collaboration**. Policies with these features have shown promising effects in addressing cybercrime. To support policymakers, we have developed the **Cybersecurity Policy Scoring Card (CSIPC)**, a tool designed to evaluate and refine existing policies for greater effectiveness.

2. Cybercrime Prediction Model

The number of network servers and public education spending significantly impact cybercrime volume, while GDP and population size influence losses. To reduce losses, policies that strengthen economic security are favored. To lower attack frequency, policies should enhance internet security and public education.

3. Conclusion

Our research offers valuable tools for cybersecurity policy development and risk prediction. Given the global nature of cyber threats, we hope our findings contribute to a safer digital future and foster international cooperation in cybersecurity efforts.

Thank you for your attention. We look forward to potential collaborations to build a secure and resilient cyberspace.

Yours Sincerely,



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Memorandum

To: Country Leaders attending the ITU Summit on Cybersecurity

From: Team # 2518848

Date: January 28th, 2025

Subject: Key Legal Strategies for Effective National Cybersecurity Policies

Cybersecurity is a critical pillar of national security and economic stability in our interconnected world. Therefore, Our research analyzed global cybercrime patterns, policy effectiveness, and socio-economic factors to identify actionable strategies for improving cybersecurity governance.

Key Findings

1. Core Drivers of Cybersecurity Success:

Countries with **higher education levels and broader internet adoption** experience fewer successful cyberattacks and faster incident response; **economic strength (GDP)** plays a secondary role, providing resources but not guaranteeing security outcomes.

2. Legal Frameworks Matter:

Nations with **comprehensive cybersecurity laws** (e.g., Canada) excel in protecting critical infrastructure, privacy, and data. Conversely, countries lacking robust legislation (e.g., Honduras) face heightened vulnerabilities.

3. Targeted Policy Reduces Specific Crimes:

For example, the U.S. emphasis on **technical safeguards and strict sentencing** has reduced phishing and payment fraud rates below global averages.

Recommendations

1. Prioritize Digital Inclusion and Education:

Expand internet access and invest in cybersecurity literacy programs to empower citizens and businesses to recognize and mitigate threats.

2. Build Comprehensive Legal Systems:

Enact laws addressing all stages of cybercrime: prevention, detection, prosecution, and recovery; Strengthen data privacy regulations and government oversight of critical infrastructure.

3. Learn from Global Peers:

Collaborate with high-performing countries to adopt best practices, particularly in privacy protection and infrastructure resilience.

Cybersecurity is a collective responsibility, shaped by education, technology, and governance. Prioritizing digital inclusion, enacting robust legal frameworks, and fostering international cooperation are crucial to mitigating cyber risks and securing digital futures.

Respectfully submitted,



<memo>-S

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Memo

TO: The International Council of Plastic Waste Management (ICM)

FROM:

DATE: January 28th, 2025

SUBJECT: Network security work

Dear sir or madam:

With the rapid development of the global Internet, the network economy has risen rapidly, and the network has connected the whole world, but it has also brought severe challenges. Cybercrime is becoming more and more rampant. Many institutions, departments and enterprises suffer from network intrusion, and important information is leaked, resulting in serious economic losses and national security threats. With the acceleration of digital transformation, network security has become an important part of national security. Our research aims to use data to show the factors that are most associated with cybercrime in various countries.

Our advantages:

- We choose factors from both positive and negative aspects.
- We scored the policy quantitatively.
- We conducted an accurate qualitative analysis and verified the results.

Here are our final results:

The most influential factor for GCI is **organizational measures, followed by capacity development.**

We should strengthen the implementation and testing of organizational policies to ensure the effective implementation of policies and play a role. At the same time, we should also strengthen the cultivation of people's ability level, carry out a series of training activities on the importance of national security and the serious harm of cybercrime, and improve people's awareness of network security. Only by raising the level of network security awareness of the public around the world can we reduce cybercrime incidents from the root and improve GCI.